Statistical Modelling

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Outline

Course Content

Course Material

Structure of the course and final assessment

Statistical Science

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Master in Data Science: Statistical Modelling, theory and applications with R

- *E-learning page* `https://elearning.unimib.it/course/view.php?id=57936`
- *Timetable* of the course
- Classroom at *lab* 1401 and 14A1: each Monday, Wednesday, and Friday (9 weeks)
- *Office hours* of the Professor and tutor are announced on the e-learning page
- *Syllabus* at `https://elearning.unimib.it/course/view.php?id=61969`

Course Overview

Main Focus

Advanced statistical modelling and reproducible research using R

- Multiple Linear Regression
- Generalized Linear Models
- Finite Gaussian Mixture Models
- Prediction
- Reproducible research with RMarkdown

Learning Objectives

- Understand theoretical foundations of advanced statistical models
- Use of the estimation methods
- Evaluate model assumptions critically
- Develop reproducible analytical workflows

Core Statistical models and estimation methods

- Univariate and multivariate random variables, univariate, bivariate and multivariate Gaussian distribution
- Multiple Linear Regression
- Ordinary Least Squares (OLS)
- Maximum Likelihood Estimation (MLE)
- Bootstrap Resampling
- Logistic Regression

Model diagnostics

- *Residual analysis*: outliers, leverage and influential points
- *Model selection* using information criteria
- *Prediction* and performance evaluation
- *Performance Metrics*: ROC Curve, AUC, Classification Error, Brier Score

Model-based clustering

- *Finite Gaussian mixture models*
- *Expectation-Maximisation (EM) Algorithm*
- *Supervised probabilistic classification*

Software and Reproducibility

- The course provides *skills* in using the semantics of the software environment R for descriptive univariate and multivariate data analysis, dimensionality reduction, and the proposed inferential methods
- R and RStudio, RMarkdown and knitr
- The aim is to create *live code, output, and comments on the results in the same interface* and to produce *reproducible documents with comments on the results* in order to acquire skills in favor of reproducibility and replicability

Key Competence

Production of *reproducible* and *replicable* reports

Applications

Applications use real and simulated data related to:

- Economics and Finance
- Social Sciences
- Biology and Ecology
- Environmental Sciences

Key Skills Developed

Outcomes

- *Critical thinking in statistical modelling*
- *Theoretical knowledge and applied data analysis skills*
- *Independent learning capability*
- *Scientific communication skills*

Prerequisites

- Probability and Statistical Inference
- Linear Algebra basics
- Discrete and Continuous Distributions
- Basic R programming knowledge

Course Material

- *Course material* is made of:
  - *slides*
  - *teaching notes*
  - *notes on the applicative example* including *data*, *R code* and comments,
  - additional material
  - *Exercises and solutions are also provided weekly*
- This material is updated at the *end* of each lecture on the e-learning page in the defined sections
- The *recorded lecture* will be made available upon successful registration.

Structure of the course

- The course is composed of *theoretical lessons and applied practical examples and case studies carried out with the software*
- A *tutor* (Dr. Brusa) will provide some lectures to illustrate the exercises and check and discuss your R assignments, lectures with the tutor are marked in the calendar published on the e-learning page
- Students are requested to *bring their laptops* or to use the pc in the lab with the *link* to the virtual lab
- The code for completing the exercises is provided in the application handouts. However, students *are free to choose which code to use* when solving the exercises. Those who implement their own code must install the required libraries in their R session in the virtual lab

Final exam

- Students can earn *two bonus points*. One point will be awarded by submitting an assignment through the Learning platform, and another point will be awarded by participating in a lab session
- The bonus will count toward the final grade of 30
- On the e-learning page an advice will announce the exercises to return and the dates
- The exam will be *open book*, therefore students are required to prepare a folder to download from the personal Gdrive containing all the material you want to use before the exam
- A first assessment of the proficiency is scheduled for the second week of *May*

Final exam

- The *final grade* will be with a final written exam that is made of *practical data analysis* and theoretical questions similar to those assigned for the homework to solve with R in RStudio using RMarkdown
- The exam is *open book and open notes and code*
- It is required to write a peproducible report showing code, results and *comments*
- It will last up to 1 hour and 30 minutes (with abut 10 questions)

Final exam

- *Fac-simile* of the exam type is provided on the e-learning page
- You must return *two files*: one with an extension .Rmd (source file) and the other a pdf file produced from the source file
- *Optional oral exam* dealing with all the program, theoretical questions and formulas, comments on the code and results
- Example of resolution

Statistical Modelling

- This course does not cover basic probability, inferential statistical aspects and matrix algebra concepts, see also the short introduction which will be provided in the elearning page

- A suggested reference for some *advanced topics in probability and statistics is the following*
  `https://mason.gmu.edu/~jgentle/books/MathStat.pdf`

- A suggested reference for some *basic topics of the R syntax* is the following `https://www.datacamp.com/courses/free-introduction-to-r`

Questions?

Statistics: Suggested books and references

- *Teaching notes* contain many useful references, many suggested books are available at the *library*
- Main *theoretical textbooks* that are available at the library are the following, other references related to the topics covered in the course will be provided in the teaching notes
- Students interested to further develop topics covered during the course think to carried out *master thesis* with me once passed the exam
- We expect a *regular attendance* to the course mainly if you wish to do the first assessment in May
- *We do not allow students to participate in streaming*; we will try to record some lectures and tutoring sessions, but we cannot guarantee the quality and timing of the recordings

Primary References (1)

- Agresti, A. and Kateri, M. (2021). *Foundations of Statistics for Data Scientists: With R and Python*. Chapman and Hall/CRC, Boca Raton.
- Bishop, Y. M., Fienberg, S. E., Holland, P. W. (2007). *Discrete Multivariate Analysis: Theory and Practice*. Springer.
- Bouveyron, C., Celeux, G., Murphy, T. B., Raftery, A. E. (2019). *Model-Based Clustering and Classification for Data Science*. Cambridge University Press.
- Fahrmeir, L., Kneib, T., Lang, S., Marx, B. D. (2021). *Regression: Models, Methods and Applications*. Springer.

Primary References (2)

- Faraway, J. J. (2014). *Extending the Linear Models with R* (2nd ed.). Chapman & Hall/CRC.
- Hastie, T., Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
- McCullagh, P., Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman & Hall/CRC.
- Pennoni, F. (2026). *Handout of Statistical Modelling: Theory and Applications with R*.
- Xie, Y., Dervieux, C., Riederer, E. (2020). *R Markdown Cookbook*. Chapman & Hall/CRC.

Statistical Science

- Statistics is defined as a set of *theoretical frameworks and methods* that allow the study of *collective phenomena*, and it can be described as the *science of learning from experience*

- It is a science having almost 250-year *history*

- *Persi Diaconis* wrote:
  "*Statistics is as healthy as it's ever been. One can see the prominence of machine learning, but they are really just using ideas that were developed in Statistics twenty or fifty years ago. They are applying them— that's great—but we are inventing the ideas that will be applied in the next twenty or fifty years. Statistics is a great field to be part of, and I'm still excited by it.*"

- "*It is the job of statistical inference to connect "dangling algorithms" to the central core of well-understood methodology. . . . , Our optimistic scenario has the big-data/data-science prediction world rejoining the mainstream of statistical inference, to the benefit of both branches.*"

- Statistical data analysis has various *goals* that involve the following aspects, among others:

  1. Understanding data through *summary* statistics and *visualizations*;
  2. Explaining the observed associations between variables through suitable *measures*;
  3. Modelling one or more variables through the others through *statistical models*;
  4. Establishing *causal* claims through the theory of *potential outcomes*
  5. Using the estimated models to make *predictions* over the phenomenon of interest;

ChatGPT's opinion

- February 1st 2025:

Do you think statistics will be useful in the future, or will only machine learning matter?

Risposta:

1. **Machine learning is based on statistics**
   Many ML algorithms are derived from statistical concepts, such as regression, Bayesian inference, and probability distributions. Without a strong statistical foundation, it becomes difficult to interpret models and their results.
2. **Understanding and interpretability**
   Statistics help explain *why* a machine learning model produces certain results. This is crucial in fields like medicine, finance, and law, where decisions must be justified.

ChatGPT's opinion

- February 1st 2025

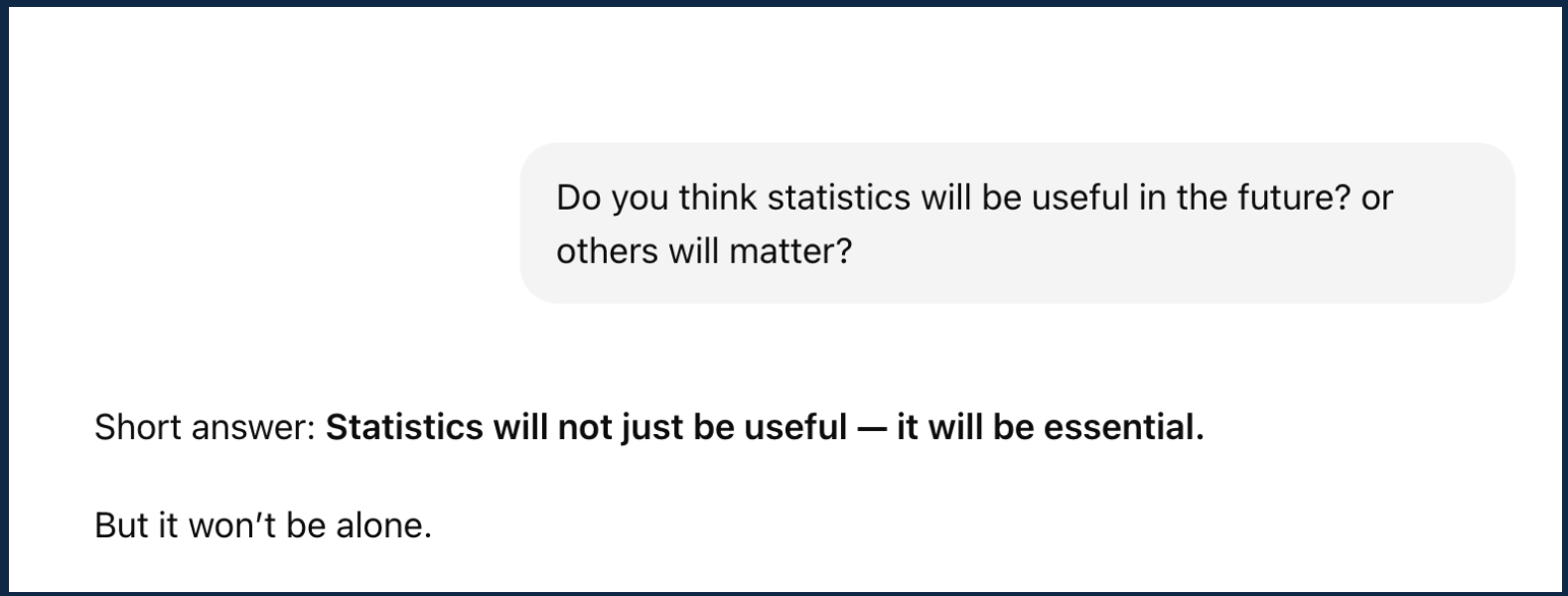
3. **Model validation**
   To verify whether a machine learning model is valid, statistical tools like hypothesis testing, p-values, and confidence intervals are needed. Without them, there is a risk of applying incorrect or overfitted models.
4. **Data quality**
   Machine learning relies on clean and representative data. Statistics provide methods to detect anomalies, handle missing data, and ensure that analyses are reliable.
5. **Machine learning is not a universal solution**
   While machine learning is powerful, it is not always necessary. In many cases, simpler statistical methods (e.g., a well-executed linear regression) are more effective and interpretable than a complex neural network.

Conclusion:

Statistics will never be replaced by machine learning but will always complement it. Anyone working with AI and data must master both fields! 🚀

ChatGPT's opinion

- March 1st 2026:



ChatGPT's opinion

Why Statistics Will Matter Even More

We live in a world driven by:

- 📊 Data (massive and messy)
- 🤖 AI and machine learning
- 🌍 Complex global systems (climate, health, economics)
- 🧬 Scientific innovation

All of these rely on **statistical thinking**, not just computation.

Statistics provides:

- Understanding of uncertainty
- Causal reasoning
- Model validation
- Critical evaluation of evidence
- Decision-making under risk

AI without statistics is just automation without understanding.

ChatGPT's opinion

The Real Future Skill

Not "statistics alone", but:

> **Statistical thinking + computational skills + critical reasoning**

The future belongs to people who can:

- Ask the right questions
- Model complex systems
- Interpret uncertainty
- Avoid misleading conclusions